# SURA.

## △ Hybrid Indoor Localisation

↳ Aim to create an ML model to detect location (using wifi & Mag. field) in GPS denied scenarios

NOTE :- Microsoft Indoor Localisation competition (MILC)

Previous Research:

1) Miskolc IIS Hybrid IPS Dataset – wifi & BT & Mag. (Not Available)
2) MagWi Benchmark Dataset – Wifi & Mag. (Dataset Only) (KNN → 90% < 10m)
3) A multi pronged approach
   ↳ Uses wifi + Cellular Signal (Relatively stable) + Magnetic field
   ↳ Avg Err = 2.91 m & Median Err. = 2.89 m
   (Mag. field alone has better avg. err but low accuracy) ← too bad at some places
   ⇒ CUSTOM Dataset, Not Available
4) Comparing CNN & LSTM
   ↳ Uses MagPie Dataset
   ↳ Mag. field Only
   ↳ Focus on CNN & LSTM → Uses filter to input data
   ↳ Avg. Err. = 1 – 1.3 m

* 5) DeepPositioning : Intelligent Fusion
   ↳ Mag field + wifi
   ↳ Custom Dataset (single phone)
   ↳ Avg → 1.4 m best

6) Indoor Localisation : Bi-LSTM

↳ Mag. field only

↳ Highlights accuracy of Bi-LSTM

↳ Also, uses Kalman filter

↳ Personal Dataset (N/A)

↳ Avg Error ≈ 0.02m – 0.1m

**Comparison: Why the results look so different**

| Feature | This Paper | Typical "Real World" Paper |
|---|---|---|
| **Step Interval** | **0.3 meters** | 1.0 – 2.0 meters |
| **Best RMSE** | **0.019 m** | 0.5 – 2.0 meters |
| **Filtering** | EMD (Aggressive) | Moving Average or Kalman |
| **Phone Handling** | Fixed/Navigating | Random/Mixed |

Export to Sheets

**The Verdict:** You are right to be skeptical. The dataset is extremely high-resolution and the collection was done under "ideal" conditions. While the **Bi-LSTM** model is technically impressive, the 2cm accuracy is a result of a very controlled environment and a very dense dataset, which would be difficult (and expensive) to replicate in a large-scale real-world building.

7) Mitigating Device Heterogenity ... - Deep Feature Learning

↳ wifi only

↳ CNN + LSTM works good

↳ Uses TUJI1 dataset (wifi Only, multidevice, 40-45cm gap)

↳ Avg. Error → 2.12m (KNN) + CNN + LSTM

8) Practical wifi Indoor Localisation

↳ wifi Only

↳ Uses GNNs for more accurate & robust to obstacle model

↳ Personal Dataset

↳ Avg. Error → ≈ 0.28m → spaced 0.5m

9) Accurate Localisation Method ... Hybrid Neural Networks

↳ Mag. field only

↳ Uses 5D-HLSTM & 8DBiLSTM , EKF (kalman) & EMD reduce noise & cumul... error

↳ Personal Dataset

↳ Avg Err < 0.5m but Err (θ) = 10° (Azimuth)

# INFERENCE :-

↳ The selected dataset (MagWi : Benchmark Dataset for Long Ter...) is a very reliable and complete dataset.

Unlike others which lack: i) Multiple orientations

ii) Different Phone sensors

iii) multiple Human Body Scen